

$$1. f_{XY}(x,y) = \frac{1}{\pi\sqrt{3}} \exp\left\{-\frac{2}{3}(x^2 - xy + y^2)\right\}$$

Show that X and $Z = (2Y - X)/\sqrt{3}$ are independent $N(0,1)$ random variables.

$$\text{and } P(X > 0, Y > 0) = 1/3$$

$$f_{XY}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right), 1-\rho^2 = \frac{3}{4}$$

$$\therefore \rho = 1/2$$

$$\therefore X, Y \sim N(0,1^2) \quad f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right), f_Y(y) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right)$$

$$Z = (2Y - X)/\sqrt{3} \quad \therefore Z \sim \text{Normal distribution}$$

$$E[Z] = \frac{1}{\sqrt{3}} (E[Y] \cdot 2 - E[X]) = 0$$

$$V[Z] = \frac{1}{3} V[2Y - X] = \frac{1}{3} (4V[Y] + V[X] - 4\text{Cov}(X,Y))$$

$$\text{Cov}(X,Y) = E[XY] - E[X]E[Y] = \rho = 1/2$$

$$\therefore V[Z] = \frac{1}{3} (4 + 1 - 4 \cdot \frac{1}{2}) = 1$$

$$\therefore Z \sim N(0,1^2)$$

$$\frac{1}{\sqrt{6\pi}} \frac{2}{\sqrt{3}} \sim 1$$

$$P(X > 0, Y > 0) \text{ with } Z = \frac{2Y - X}{\sqrt{3}}, Y = (X + \sqrt{3}Z)/2$$

$$P(X > 0, Y > 0) = P(X > 0, \frac{X + \sqrt{3}Z}{2} > 0) = P(X > 0, Z > -\frac{X}{\sqrt{3}})$$

$$\int_0^\infty P(Z > -\frac{X}{\sqrt{3}} | X=x) f_X(x) dx = \int_0^\infty \Phi\left(\frac{X}{\sqrt{3}}\right) f_X(x) dx = \frac{1}{4} + \frac{1}{2\pi} \arctan\left(\frac{1}{\sqrt{3}}\right)$$

$$= \frac{1}{4} + \frac{1}{2\pi} \cdot \frac{\pi}{6} = \frac{1}{3}$$

2. fixed N light bulbs. in a box $X_n \sim \text{Exp}(n)$

a) Find probability that selected bulb last more than one month

$$P(X_n = t) = ne^{-nt} \quad t > 0$$

$$P(X_n > 1) = \int_1^\infty ne^{-nt} dt = -e^{-nt} \Big|_1^\infty = \frac{1}{e^n}$$

$$P(X > 1) = P(n=1)P(X_1 > 1) + \dots = \frac{1}{N} \sum_{n=1}^\infty \frac{1}{e^n} = \frac{1}{N} \frac{1/e}{1 - 1/e}$$

$$b) P(Y=y, X>1) = \frac{P(X>1|Y=y)}{P(X>1)} = \frac{P(X_n>1)}{P(X>1)} = \frac{e^{-n}}{\frac{1}{N} \sum_{k=1}^N e^{-k}}$$

3. X and Y are jointly (bivariate) normal

$\text{Var}(X) = \text{Var}(Y)$ Show that $X+Y$ and $X-Y$ are independent

$$X \sim N(\mu_X, \sigma^2) \quad Y \sim N(\mu_Y, \sigma^2)$$

$$U = X+Y, \quad V = X-Y$$

$$\begin{aligned} \text{Cov}(U, V) &= \text{Cov}(X+Y, X-Y) = \text{Cov}(X, X) - \text{Cov}(X, Y) + \text{Cov}(Y, X) - \text{Cov}(Y, Y) \\ &= \text{Var}(X) - \text{Var}(Y) = 0 \end{aligned}$$

$\text{Cov}(U, V) = 0$ iff U, V are independent

4. X and Y are jointly normal random variables

$$\mu_X = 0, \sigma_X^2 = 1, \mu_Y = -1, \sigma_Y^2 = 4, \rho = -1/2$$

$$a) P(X+Y > 0)$$

$$X \sim N(\mu_X, \sigma_X^2) = N(0, 1^2)$$

$$\rho(X, Y) = \rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$Y \sim N(\mu_Y, \sigma_Y^2) = N(-1, 2^2)$$

$$\therefore \text{Cov}(X, Y) = \rho \cdot 1 \cdot 2 = 2\rho = -1$$

$$X+Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2 + 2\text{Cov}(X, Y))$$

$$= N(-1, 3)$$

$$P(X+Y > 0) = P(Z > \frac{0+1}{\sqrt{3}}) = P(Z > 1/\sqrt{3})$$

b) Find constant a if we know $aX+Y$, $X+2Y$ are independent

$$\text{Cov}(aX+Y, X+2Y) = a\text{Cov}(X, X) + 2a\text{Cov}(X, Y) + \text{Cov}(Y, X) + 2\text{Cov}(Y, Y)$$

$$= a\text{Var}(X) + (2a+1)\text{Cov}(X, Y) + 2\text{Var}(Y)$$

$$= a + (2a+1)(-1) + 4 = a - 2a - 1 + 4 = 7 - a = 0$$

$$\therefore a = 7$$

$$c) P(X+Y > 0 | X+Y > -3) = \frac{P(X+Y > 0)}{P(X+Y > -3)}$$

$$X+Y \sim N(-1, \sqrt{3})$$

$$\Rightarrow P(X+Y > 0 | X+Y > -3) = \frac{P(Z > \frac{1}{\sqrt{3}})}{P(Z > -\frac{2}{\sqrt{3}})} \approx 0.3218$$

5. Gubner Chapter 7. Problem 57

$$f_{X|Z}(x, y, z) = \frac{2 \exp[-|x-y| - (y-z)^2/2]}{z^5 \sqrt{2\pi}}, \quad z \geq 1$$

$$f_{X|Z}(x, y, z) = 0 \text{ otherwise}$$

$$f_X(x, y) = \int_1^\infty f_{X|Z}(x, y, z) dz = \frac{2}{\sqrt{2\pi}} \exp[-|x-y|] \int_1^\infty \frac{1}{z^5} \exp\left(-\frac{(y-z)^2}{2}\right) dz$$

$$\begin{aligned} f_{Y|Z}(y, z) &= \int_{-\infty}^\infty f_{X|Z}(x, y, z) dx = \frac{2 \exp[-(y-z)^2/2]}{\sqrt{2\pi} z^5} \int_{-\infty}^\infty \exp(-|x-y|) dx \\ \int_{-\infty}^\infty \exp(-|x-y|) dx &= \int_y^\infty \exp(-x+y) dx + \int_{-\infty}^y \exp(x-y) dx \\ &= -\exp(-x+y) \Big|_y^\infty + \exp(x-y) \Big|_{-\infty}^y = \exp(0) + \exp(0) = 2. \end{aligned}$$

$$\therefore f_{Y|Z}(y, z) = \frac{4 \exp[-(y-z)^2/2]}{z^5 \sqrt{2\pi}}$$

$$f_Z(z) = \int_{-\infty}^\infty f_{Y|Z}(y, z) dy = \frac{4}{z^5 \sqrt{2\pi}} \int_{-\infty}^\infty \exp[-(y-z)^2/2] dy$$

$$\text{with } \frac{y-z}{\sqrt{2}} = t \Rightarrow \frac{1}{\sqrt{2}} dy = dt$$

$$\int_{-\infty}^\infty \exp[-(y-z)^2/2] dy = \int_{-\infty}^\infty \exp[-t^2] \sqrt{2} dt = \sqrt{2} \cdot \sqrt{\pi} = \sqrt{2\pi}$$

$$\therefore f_Z(z) = \frac{4}{z^5 \sqrt{2\pi}} \cdot \sqrt{2\pi} = \frac{4}{z^5}$$

$$f_{X|Z}(x|y, z) = \frac{f_{X|Z}(x, y, z)}{f_{Y|Z}(y, z)} = \frac{\frac{2 \exp[-|x-y|] \exp[-(y-z)^2/2]}{z^5 \sqrt{2\pi}}}{\frac{4 \exp[-(y-z)^2/2]}{z^5 \sqrt{2\pi}}} = \frac{2 \exp[-|x-y|]}{4} = \frac{\exp[-|x-y|]}{2}$$

$$\begin{aligned} f_{Y|Z} &= \frac{f_{Y|Z}(y, z)}{f_Z(z)} = \frac{\frac{4 \exp[-(y-z)^2/2]}{z^5 \sqrt{2\pi}}}{\frac{4}{z^5}} \\ &= \frac{\exp[-(y-z)^2/2]}{\sqrt{2\pi}} \sim N(z, 1^2) \end{aligned}$$

$$E[XY] = E[E[XY|YZ]]$$

$$= \int \int y E[X|Y=y, Z=z] f_{X|Y,Z}(x|y,z) dy dz$$

$$f_{X|Y,Z}(x|y,z) = \frac{\exp[-|x-y|]}{2} \quad \text{shifted Laplace}$$

$$E[X|Y=y, Z=z] = \int_{-\infty}^{\infty} x \frac{\exp[-|x-y|]}{2} dx = \frac{1}{2} \int_{-\infty}^{\infty} x \exp[-|x-y|] dx$$

$$= \frac{1}{2} \int_y^{\infty} x \exp[-x+y] dx + \frac{1}{2} \int_{-\infty}^y x \exp[x-y] dx$$

$$\therefore E[X|Y=y, Z=z] = y$$

$$\Rightarrow E[XY] = \int_{-\infty}^{\infty} y^2 f_{X|Y,Z}(x|y,z) dy dz = E[Y^2]$$

$$E[Y^2] = E[E[Y^2|Z]] = E[1+Z^2] = \int_1^{\infty} (1+z^2) \frac{4}{z^5} dz$$

$$= \int_1^{\infty} \frac{4}{z^5} dz + \int_1^{\infty} \frac{4}{z^3} dz = -z^{-4} \Big|_1^{\infty} - 2z^{-2} \Big|_1^{\infty} = 1+2=3$$

$$E[YZ] = E[E[YZ|Z]] = E[Z E[Y|Z]] = E[Z^2]$$

$$= \int_1^{\infty} z^2 \frac{4}{z^5} dz = \int_1^{\infty} \frac{4}{z^3} dz = 2$$

6. Gubner Chapter 8. Problem 10.

$$U \sim N(0,1^2), \quad X = ZU + Y, \quad f_Z(z) = \frac{3}{4} z^2 \quad 1 \leq z \leq 2$$

$$Y|Z=z \sim \text{Exp}(z)$$

$$E[Z] = \int_1^2 \frac{3}{4} z^2 \cdot z dz = \frac{3}{4} \left[\frac{1}{4} z^4 \right]_1^2 = \frac{3}{28} (16-1) = 45/28$$

$$E[Y] = E[E[Y|Z]] = E[1/Z] = \int_1^2 \frac{3}{4} z^2 \cdot \frac{1}{z} dz = \frac{3}{4} \int_1^2 z dz = \frac{3}{4} \cdot \left[\frac{1}{2} z^2 \right]_1^2 = \frac{3}{4} \cdot \frac{1}{2} (4-1) = \frac{9}{14}$$

$$E[X] = E[ZU] + E[Y] = E[Z]E[U] + E[Y] = \frac{45}{28} \cdot 0 + \frac{9}{14} = \frac{9}{14}$$

mean vector $\begin{pmatrix} E[X] \\ E[Y] \\ E[Z] \end{pmatrix} = \begin{pmatrix} 9/14 \\ 9/14 \\ 45/28 \end{pmatrix}$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$\begin{aligned} E[XY] &= E[(ZU+Y)Y] = E[ZUY] + E[Y^2] \\ &= E[Z E[UY|Z]] + E[Y^2] \\ &= E[Z E[U|Z] E[Y|Z]] + E[E[Y^2|Z]] \\ &= E[Z \cdot 0 \cdot \frac{1}{Z}] + E[\frac{1}{Z^2} + \frac{1}{Z^2}] = 2E[\frac{1}{Z^2}] = \frac{6}{7} \end{aligned}$$

$$\therefore \text{Cov}(X, Y) = \frac{6}{7} - \frac{9}{14} \cdot \frac{9}{14} = \frac{168 - 81}{196} = \frac{87}{196}$$

$$\text{Cov}(Y, Z) = E[YZ] - E[Y]E[Z]$$

$$\begin{aligned} &= E[Z E[Y|Z]] - E[Y]E[Z] = E[Z \cdot \frac{1}{Z}] - E[Y]E[Z] \\ &= 1 - \frac{9}{14} \cdot \frac{45}{28} = \frac{392 - 405}{392} = -\frac{13}{392} \end{aligned}$$

$$\text{Cov}(Z, X) = E[ZX] - E[Z]E[X]$$

$$\begin{aligned} &= E[Z(ZU+Y)] - E[Z]E[X] = E[Z^2U] + E[YZ] - E[X]E[Z] \\ &= E[Z^2]E[U] + E[YZ] - E[X]E[Z] = 1 - \frac{9}{14} \cdot \frac{45}{28} = -\frac{13}{392} \end{aligned}$$

$$\text{Var}(X) = \text{Var}(ZU) + \text{Var}(Y) + 2\text{Cov}(ZU, Y)$$

$$\text{Cov}(ZU, Y) = E[ZUY] - E[Z]E[U]E[Y] = 0$$

$$\begin{aligned} \therefore \text{Var}(X) &= \text{Var}(ZU) + \text{Var}(Y) = E[Z^2U^2] - E[Z]^2E[U]^2 + \text{Var}(Y) \\ &= E[Z^2] + \text{Var}(Y) \end{aligned}$$

$$\begin{aligned} \text{Var}(Y) &= E[Y^2] - E[Y]^2 = E[E[Y^2|X]] - E[Y]^2 = E\left[\frac{2}{Z^2}\right] - E[Y]^2 \\ &= \int_1^2 \frac{6}{7} dz - E[Y]^2 = \frac{6}{7} - \left(\frac{9}{14}\right)^2 = \frac{168 - 81}{196} = \frac{87}{196} \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Var}(X) &= \int_1^2 \frac{2}{7} z^4 dz + \text{Var}(Y) = \frac{2}{7} \cdot \frac{1}{5} z^5 \Big|_1^2 + \text{Var}(Y) = \frac{3 \cdot 31}{35} + \frac{87}{196} \\ &= \frac{3039}{980} \end{aligned}$$

$$\text{Var}(Z) = E[Z^2] - E[Z]^2 = \frac{93}{35} - \left(\frac{45}{28}\right)^2 = \frac{93}{35} - \frac{2025}{784}$$

Covariance Matrix of $[X \ Y \ Z]$

$$\begin{pmatrix} \text{Var}(X) & \text{Cov}(X,Y) & \text{Cov}(X,Z) \\ \text{Cov}(X,Y) & \text{Var}(Y) & \text{Cov}(Y,Z) \\ \text{Cov}(X,Z) & \text{Cov}(Y,Z) & \text{Var}(Z) \end{pmatrix}$$

7. Gubner Chapter 8, Problem 11.

$$f_{XYZ}(x,y,z) = \frac{2 \exp[-|x-y| - (y-z)^2/2]}{z^5 \sqrt{2\pi}}, \quad z \geq 1$$

from Q5

$$f_Z(z) = \frac{4}{z^5}$$

$$f_{X|YZ}(x|y,z) = \frac{\exp[-|x-y|]}{2}$$

$$f_{Y|Z}(y|z) = \frac{\exp[-(y-z)^2/2]}{\sqrt{2\pi}} \sim N(z, 1)$$

$$\begin{aligned} E[X] &= E[E[X|Y,Z]] = E[Y] = E[E[Y|Z]] - E[Z] \\ &= \int_1^\infty z \cdot \frac{4}{z^5} dz = 4 \int_1^\infty \frac{1}{z^4} dz = 4 \cdot \left(-\frac{1}{3} \frac{1}{z^3}\right) \Big|_1^\infty = \frac{4}{3} \end{aligned}$$

mean vector $\begin{pmatrix} 4/3 \\ 4/3 \\ 4/3 \end{pmatrix}$

$$E[X^2] = E[E[X^2|Y,Z]] = E[2 + Y^2] = E[Y^2] + 2$$

$$E[Y^2] = E[E[Y^2|Z]] = E[1 + Z^2] = 1 + E[Z^2]$$

$$E[Z^2] = \int_1^\infty z^2 \cdot \frac{4}{z^5} dz = 4 \left(-\frac{1}{2} \frac{1}{z^3}\right) \Big|_1^\infty = 2$$

$$\therefore E[Y^2] = 1 + 2 = 3$$

$$E[X^2] = 3 + 2 = 5$$

$$\text{Var}(X) = 5 - \frac{16}{9} = \frac{29}{9}$$

$$\text{Var}(Y) = 3 - \frac{16}{9} = \frac{11}{9}$$

$$\text{Var}(Z) = 2 - \frac{16}{9} = \frac{2}{9}$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = E[Y^2] - E[X]E[Y]$$

$$= 3 - \frac{16}{9} = \frac{11}{9}$$

$$\text{Cov}(Y, Z) = E[YZ] - E[Y]E[Z] = E[Z^2] - E[Y]E[Z]$$

$$= 2 - \frac{16}{9} = \frac{2}{9}$$

$$\text{Cov}(Z, X) = E[XZ] - E[X]E[Z]$$

$$= E[Z E[X|Y, Z]] - E[X]E[Z] = E[YZ] - E[X]E[Z]$$

$$= E[Z^2] - E[X]E[Z] = 2 - \frac{16}{9} = \frac{2}{9}$$